

Module 05: Action — Robot Movers

Learning Objectives

1. Understand that **Actuators** (Motors) make robots move.
2. Learn the difference between **Wheels** (Rolling) and **Legs** (Walking).
3. Know that every action uses **Power** (Battery).

Introduction: Robot Movers

How do you move? You use muscles. How does a robot move? It uses **Actuators**!

- **Motors**: Spin round and round (like a fan).
- **Pistons**: Push in and out (like a pump).

Without actuators, a robot is just a smart rock. It can think, but it can't DO anything!

Key Concepts

1. Wheels vs. Legs

- **Wheels**: Fast on flat ground. Easy to build. (Cars).
 - **Legs**: Slow but can climb rocks and stairs. Hard to build! (Boston Dynamics Spot).
- Which one is better? It depends on where you are going!

2. Grippers (Hands)

Robots need to pick things up. They use **Grippers**. Some look like clamps. Some look like suction cups. They are the robot's hands.

3. Precision

Robots can move very perfectly. A surgeon robot can sew a grape! A painting robot calculates every spray. This is called **Precision**.

Activities

Activity 1: The Human Crane

Work with a partner. Partner is the **Crane**. You are the **Controller**. Tell them how to move their arm to pick up a pencil. "Lower... Lower... Close Gripper (Hand)... Lift!" Robots need simple steps to move.

Activity 2: Wheel or Leg?

Look at different terrains (Grass, Road, Stairs, Mud). Decide: Would you use Wheels or Legs?

- Road -> Wheels (Fast!).
- Stairs -> Legs (Wheels get stuck!).

Summary

Action is how robots change the world. By using motors, wheels, and grippers, they can travel to Mars, build cars, or clean your floor!

References

- *Balloons Over Broadway* (About puppeteer Tony Sarg - Mechanics of movement)
- *Mechanimals* by Chris Tougas